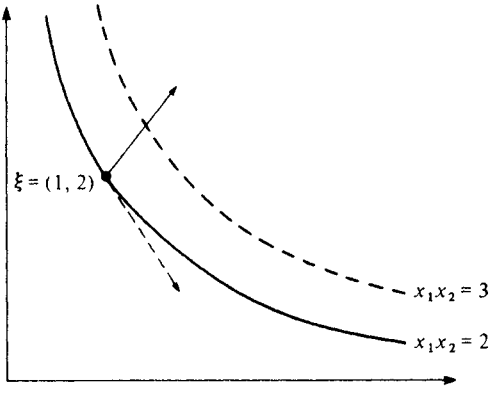
the direction (3, 4). The broken arrow indicates the direction $(1, -2)$ which is tangent to the contour $x_1x_2 = 2$ (why?). It is instructive to check that the directional derivative at (1, 2) in the direction $(1, -2)$ is zero.



19.3 Partial derivatives

Suppose that f: $\mathbb{R}^n \to \mathbb{R}$. To avoid cumbersome notation we shall assume in what follows that $n = 2$ but the general case is entirely analogous.

If $\boldsymbol{\xi} \in \mathbb{R}^2$ the *partial derivatives* of f at $\boldsymbol{\xi}$ are defined by

$$f_1(\xi_1, \xi_2) = \lim_{h \to 0} \frac{f(\xi_1 + h, \xi_2) - f(\xi_1, \xi_2)}{h}$$

$$f_2(\xi_1, \xi_2) = \lim_{k \to 0} \frac{f(\xi_1, \xi_2 + k) - f(\xi_1, \xi_2)}{k}$$

provided that these limits exist.

It is no more difficult to calculate the partial derivatives of a function of a vector variable than it is to calculate the derivative of a function of a real variable. To calculate f_1, for example, one simply differentiates with respect to x_1 treating x_2 as a constant.

19.4 Examples

(i) Let f: $\mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1{}^2x_2 + x_2{}^3x_1$. Then

$$\left.\begin{aligned} f_1(x_1, x_2) &= 2x_1x_2 + x_2{}^3 \\ f_2(x_1, x_2) &= x_1{}^2 + 3x_2{}^2x_1. \end{aligned}\right\}$$

(ii) Let g: $\mathbb{R}^2 \to \mathbb{R}$ be defined by $g(x_1, x_2) = \sin(x_1x_2 + x_2^2)$. Then

$$\left.\begin{aligned} g_1(x_1, x_2) &= x_2 \cos(x_1x_2 + x_2^2) \\ g_2(x_1, x_2) &= (x_1 + 2x_2)\cos(x_1x_2 + x_2^2). \end{aligned}\right\}$$

(iii) Let h: $\mathbb{R}^3 \to \mathbb{R}$ be defined by $h(x_1, x_2, x_3) = x_1 + x_2x_3$. Then

$$\left.\begin{aligned} h_1(x_1, x_2, x_3) &= 1 \\ h_2(x_1, x_2, x_3) &= x_3 \\ h_3(x_1, x_2, x_3) &= x_2. \end{aligned}\right\}$$

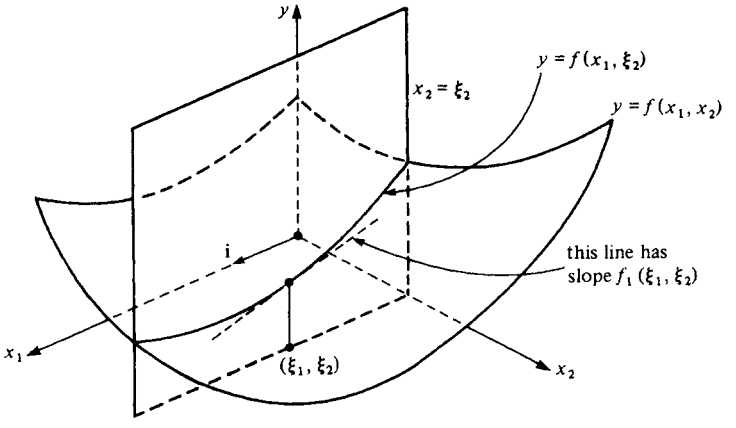
The geometric interpretation of a partial derivative is important. Let $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$. Observe that $\mathbf{i}$ is a unit vector which points in the direction of the x_1-axis while $\mathbf{j}$ is a unit vector which points in the direction of the x_2-axis. We have that

$$\begin{aligned} D_{\mathbf{i}}f(\boldsymbol{\xi}) &= \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{i}) - f(\boldsymbol{\xi})}{t} \\ &= \lim_{t\to 0} \frac{f(\xi_1 + t, \xi_2) - f(\xi_1, \xi_2)}{t} \\ &= f_1(\xi_1, \xi_2). \end{aligned}$$

Similarly,

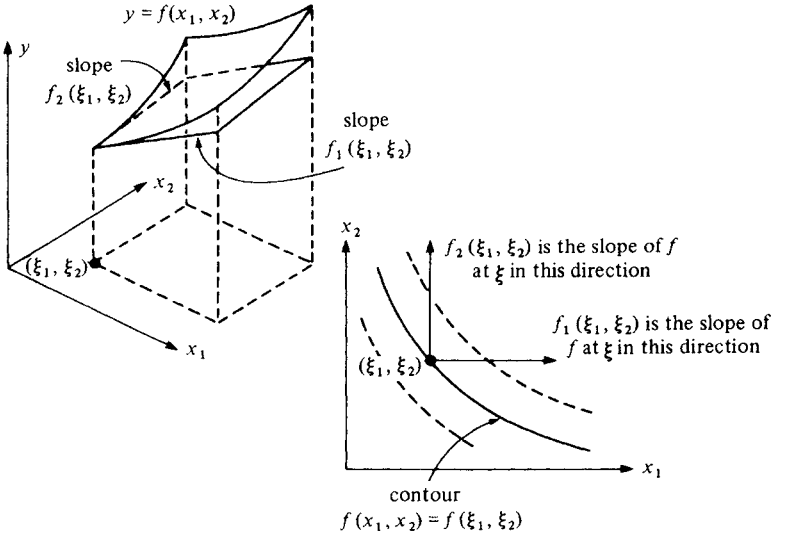
$$\begin{aligned} D_{\mathbf{j}}f(\boldsymbol{\xi}) &= \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{j}) - f(\boldsymbol{\xi})}{t} \\ &= \lim_{t\to 0} \frac{f(\xi_1, \xi_2 + t) - f(\xi_1, \xi_2)}{t} \\ &= f_2(\xi_1, \xi_2). \end{aligned}$$

Thus the partial derivative $f_1(\xi_1, \xi_2)$ is the slope of the surface $y = f(x_1, x_2)$ at the point $\boldsymbol{\xi}$ in the direction of the x_1-axis while $f_2(\xi_1, \xi_2)$ is the slope in the direction of the x_2-axis.



The diagram above illustrates the geometric significance of the partial derivative $f_1(\xi_1, \xi_2)$. It shows the surface $y = f(x_1, x_2)$ sliced by the vertical plane $x_2 = \xi_2$ (which passes through (ξ_1, ξ_2) and is parallel to $\mathbf{i} = (1, 0)$). On the plane $x_2 = \xi_2$, the graph $y = f(x_1, \xi_2)$ can be seen. The partial derivative $f_1(\xi_1, \xi_2)$ is simply the slope of this graph at the point $x_1 = \xi_1$.

The diagrams below illustrate both partial derivatives simultaneously.



19.5 Notation

As in the case of ordinary derivatives, there are numerous alternative notations which have to be kept in mind when working with partial derivatives. Which is used will depend on the context.

We note first that it is common to denote the partial derivative $f_1(\xi_1, \xi_2)$ by

$$f_{x_1}(\xi_1, \xi_2)$$

with a similar convention for $f_2(\xi_1, \xi_2)$. This may seem a clumsy piece of notation but it is quite useful when, as often happens, it is convenient to write x rather than x_1 and y rather than x_2.

However, the most important alternative notation is the use of

$$\frac{\partial y}{\partial x_1}$$

for the partial derivative $f_1(x_1, x_2)$ in the case when $y = f(x_1, x_2)$. Similarly

$$\frac{\partial y}{\partial x_2}$$

is used for the partial derivative $f_2(x_1, x_2)$. As noted under similar circumstances in §10.4, this notation suffers from the disadvantage that x_1 and x_2 are used ambiguously both as the co-ordinates of the point at which the partial derivative is to be evaluated and as the variables with respect to which one is to differentiate. This ambiguity is often irrelevant in practical applications but can be very confusing when theorems are to be proved.

When a number of variables are in play, it is not always obvious which are to be held constant when differentiating partially. But it is of the *greatest* importance to be clear on this point. Where necessary, the notation

$$\left(\frac{\partial y}{\partial x_1}\right)_{x_2}$$

is available. This means 'differentiate partially with respect to x_1 treating x_2 as constant'.

19.6 *Example* Consider the equations

$$\left.\begin{aligned} u &= x + y \\ v &= x - y. \end{aligned}\right\}$$

From the first equation

$$\frac{\partial u}{\partial x} = 1$$

If the equations are solved for x and y in terms of u and v, we obtain that

$$\left.\begin{aligned} x &= \tfrac{1}{2}(u + v) \\ y &= \tfrac{1}{2}(u - v) \end{aligned}\right\}$$